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**Understanding the Electrode/Electrolyte interface Layer on Li-rich Nickel Manganese
Cobalt Layered Oxide Cathode by XPS**

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X-ray photoelectron spectroscopy, lithium-ion battery

Abstract

Layered lithium rich nickel manganese cobalt oxide (LR-NMC) represents one of the most promising cathode materials for application in high energy density lithium-ion batteries. The extraordinary capacity delivered derives from a combination of both cationic and anionic redox processes. However, the latter ones lead to oxygen evolution which triggers structural degradation and electrode/electrolyte interface (EEI) instability that hinders the use of LR-NMC in practical application.

In this work, we investigate the surface chemistry of LR-NMC and its evolution upon different conditions to give further insights into the processes occurring at the EEI. X-ray photoelectron spectroscopy (XPS) studies reveals that once the organic component of the layer is formed, it remains stable independently on the higher cut-off voltage applied, while continuous growth of inorganics along with oxygen evolution occurs.

The results performed on lithiated and de-lithiated LR-NMC surfaces indicate an instability of the EEI layer formed at high voltages, which undergoes a partial decomposition. Furthermore, the tris(pentafluorophenyl)borane (TPFPB) electrolyte additive simultaneously prevents excess LiF formation and changes the chemical composition of the EEI layer. This latter is characterized by a higher amount of poly(ethylene oxide) oligomer (PEO) species and $\text{Li}_x\text{PO}_y\text{F}_z$ formation. In addition, the presence of boron containing compounds in the EEI layer cannot be excluded, which may be also responsible of the increased thickness of the EEI layer. Finally, fast kinetics at elevated temperatures exacerbate the salt decomposition which results in the formation of an EEI which is thicker and richer in LiF.

1. Introduction

Lithium-ion batteries (LIBs) represent the leading technology for powering portable electronics and new generation of vehicles including electric (EV), hybrid (HEV) and plug-in hybrid (PHEVs) vehicles.^{1,2} Although LIBs are already successfully commercialized, there is still room for improvement in critical aspects such as their low- and high-temperature performance, safety features, and cost.³

The steep decrease in price of LIBs observed during the last years has been certainly driven by the technological improvements and the wide competition among manufacturers.⁴ However, in view of the increasingly growing demand for LIBs in the electric automotive

field and possible use for large scale energy storage, the cost of LIBs still requires to be substantially decreased. The component mainly impacting the final cost of the battery pack is the cathode active material.^{4,5} The main factor determining the final costs is certainly the cost of the employed raw materials, which accounts for more than 50% of the total cost, which nowadays is further aggravated by the lack of long-term fixed-price contracts with suppliers of critical constituent metals such as cobalt⁶ and lithium.⁷

Beside the cost, the cathode material strongly affects the final electrochemical performance of the battery. Specific capacity and energy density depend on the cathode material employed. Indeed, the lithium ion storage ability in positive electrodes affects the overall delivered capacity and the operating cell voltage determines the final energy density.⁸ Therefore, establishing desirable cathodes is one of the greatest concerns and challenges for the development of future high-performance LIBs. 4 V spinel oxides (LiMn_2O_4) and 3.4 V olivine (LiFePO_4) cathodes have been used owing to their environmentally friendly nature, non-toxicity, low-cost, thermal stability and superior cycle life performance.⁹ Both materials exhibit distinctive cost advantages when compared to LiCoO_2 , and their structures are more stable to lithium extraction, but they do not offer any significant advantage in terms of energy density.

Thackeray *et al.* proposed an elegant strategy to increase the stability and improve the electrochemical behaviour of lithium metal layered oxides (LiMO_2 , $\text{M} = \text{Mn, Ni, Co}$). The strategy consisted in the integration of structurally compatible units of Li_2MnO_3 (in layered notation, $\text{Li}[\text{Li}_{0.33}\text{Mn}_{0.67}]\text{O}_2$) as partial substituents in the LiMO_2 component^{10,11} (rocksalt-type structure).

The excess Li ions incorporated in the transition metal layers, overcomes the limit of one Li ion per MO_2 formula, thus enabling the achievement of exceptionally high reversible

capacities and energy densities when cycled between 2.0 and 4.8 V. This class of cathode materials, generally called lithium rich layered oxides hereafter abbreviated as LR-NMC are represented by the general formula $(x\text{LiMO}_2 \cdot (1-x)\text{Li}_2\text{MnO}_3)$, $M = \text{Ni, Co, Mn}$. They represent one of the most promising classes of cathode materials for the next generation LIBs. They can deliver discharge capacities larger than 250 mA h g^{-1} after being activated during the first de-lithiation process. Indeed, two distinctive regions are exhibited in the voltage profile during the first charging process. The first sloping region (up to 4.4 V) is associated with the Li extraction from the layered LiMO_2 structure and the oxidation of the nickel and cobalt transition metal ions into the tetravalent state. The second region observed at potentials higher than 4.5 V is associated with the electrochemical activation of the Li_2MnO_3 structural unit accompanied by a well-defined plateau involving lithium extraction. This process occurs simultaneously to release and following oxidation of oxygen and concomitant lithium-proton exchange processes.^{10,12–15}

The origin of the extra capacity delivered in the elongated plateau at high voltage values is still heavily discussed, however it is most likely associated with irreversible oxygen loss and/or oxidation of oxygen anions upon de-lithiation.^{9,16,17} The anionic redox reactions taking place and the consequent processes generate highly reactive species such as oxygen and peroxides,^{16,18} which react with the electrolyte and contribute to the formation of the electrode/electrolyte interface (EEI) film.^{19–22} The reactivity of layered oxides exhibiting anionic redox toward the electrolyte is still not well understood, however Yabuuchi *et al.*²³ proposed that charging to high voltages can trigger nucleophilic attacks of carbonate solvents by superoxide (O_2^-), which can form upon discharge to 3 V and lower, leading to the formation of EEI films.

Indeed, the release of oxygen from Li-rich materials has been shown to catalyse electrolyte decomposition. While studying lithium-air systems, Aurbach *et al.* proposed that superoxide species can attack carbonate solvents to form inorganic species such as Li_2CO_3 .²⁴ K. Kang *et al.* reported²⁵ for instance the partially reversible formation and decomposition of Li_2CO_3 by using a $\text{Li}[\text{Li}_{0.2}\text{Ni}_{0.2}\text{Mn}_{0.6}]\text{O}_2$ (Or $0.5\text{LiNi}_{0.5}\text{Mn}_{0.5}\text{O}_2 \cdot 0.5\text{Li}_2\text{MnO}_3$) cathode. They proposed that while the reversible Li_2CO_3 formation can contribute to the high capacity of this class of materials, the H_2O by-product resulting from the decomposition of carbonate-based electrolytes can easily hydrolyse the electrolyte salt (LiPF_6) to generate HF, LiF, and POF_3 . The acidic species attack metal oxide electrodes and lead to surface damage that deteriorates the electrochemical performance of the electrode.

As expected, chemical and electrochemical reactions of electrolyte with the cathode materials are exacerbated at elevated temperatures and high cut-off voltages.²⁶

Despite several studies have been already devoted to the investigation of the reaction mechanisms involving oxygen evolution of Li rich manganese layered oxides,^{12,15,19,21–23,27–29} the effect of the oxygen release toward the EEI layer is still not fully understood. X-Ray photoemission spectroscopy (XPS) has proven to be a powerful tool to investigate the composition of surface species. In this work, the surface chemistry of LR-NMC is characterized by examining the chemical composition of the EEI layer before and after the Li_2MnO_3 activation by XPS. In addition, the employment of Tris(pentafluorophenyl) borane additive is evaluated in half-cell configuration and the results of an increased temperature and cut-off voltage on the kinetics of the EEI layer formation and its chemical composition are investigated and discussed providing new insights into the EEI layer composition and properties of LR-NMC cathodes.

2. Experimental

2.1. Material synthesis

LR-NMC with nominal composition $\text{Li}_{1.2}\text{Ni}_{0.16}\text{Mn}_{0.56}\text{Co}_{0.08}\text{O}_2$ was synthesized by wet chemistry route followed by solid-state reaction as already described in detail in former reports.^{30,31} Briefly, a manganese-nickel-cobalt hydroxide precursor was obtained by coprecipitation method using $\text{Mn}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$ (99%, Acros Organics), $\text{Ni}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$ (98%, Aldrich), and $\text{Co}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$ (98%, Alfa Aesar) and $\text{Li}(\text{OH}) \cdot \text{H}_2\text{O}$ (98%, Alfa Aesar). After rinsing, drying and ball milling, the precursor was annealed with stoichiometric amounts of lithium hydroxide hydrate ($\text{LiOH} \cdot \text{H}_2\text{O}$ Aldrich >98%) at 900 °C for 24 hours and finally quenched in liquid nitrogen.

The final composition has been determined by inductively coupled plasma optical emission spectrometry (ICP-OES; Ultima 2 from Horiba).

2.2. Electrode preparation and electrochemical characterization

Composite electrodes were prepared by casting a slurry composed by 85 wt% active material (LR-NMC), 10 wt% conductive agent carbon black (C-Nergy, Super C 45, Imerys) and 5 wt% Sodium Carboxymethylcellulose binder (Na-CMC, DOW Wolff Cellulosics, Walocel CRT2000PA12). Na-CMC was first dissolved in deionized water by magnetically stirring the solution for 3 hours. Addition of carbon super C 45 and LR-NMC followed. The mixture was subsequently ball milled in a planetary mill (Vario-Planetary MillPulverisette 4, Fritsch) for 3 additional hours. Resulting slurries were cast on aluminium foil with a doctor-blade coater (1133 N, OLBRICH), dried overnight at 60 °C and punched as 12 mm diameter disks. 2.5-3 mg cm^{-2} was the active material mass loading of the as prepared electrodes, while after pressing at 4 tons the final thickness was 20 μm . Electrodes were dried at 120 °C under

vacuum in a glass oven for 12 hours and transferred to an Ar-filled glove box (LabMaster, Mbraun GmbH) with oxygen and moisture level of <0.1 ppm for storage and cell assembly. Three-electrode Swagelok® T-type cells were used for the electrochemical characterization. Glass fiber disc (Whatman GF/D) soaked with 120 μL of 1 M LiPF_6 in ethylene carbonate and dimethyl carbonate (EC:DMC 1:1 %wt -LP30; UBE) electrolyte was used as the separator and sandwiched between LR-NMC cathode and metallic lithium foil (Rockwood Lithium) used as counter and reference electrode. All the potential values reported in this work refer to the Li/Li^+ redox couple. The cells were galvanostatically cycled within the 2.5 V- 4.8 V vs. Li/Li^+ potential range (unless otherwise mentioned) at a constant rate of C/20 (1C is defined as 200 mAh g^{-1}) and by using a SP-200 potentiostat (Bio-logic Science Instruments). All charge/discharge cycles were performed in climatic chambers (KBF115, Binder GmbH) at 20 °C except for the high temperature experiment which was performed at 55 °C. Beside the standard electrolyte, a 0.2 molar Tris(Pentafluoro phenyl)borane (TPFPB, 97%, Alfa Aesar) additive in the standard electrolyte (LP30) was prepared by stirring overnight the solution inside a glove box.

2.3. *Ex-situ XPS analysis*

All reference samples (carbon super C 45, LR-NMC powder and Na-CMC binder) were vacuum dried overnight prior to be examined by XPS. LR-NMC electrodes that were used in cells, i.e. either kept at open circuit voltage (OCV) or cycled), were disassembled inside the glove box and dip-washed with DMC for a few seconds to remove electrolyte residue. Drying under vacuum in the glove-box antechamber for half an hour and mounting on the XPS sample holder followed. In order to prevent air exposure and surface contamination, all

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3 samples were transferred to the XPS load-lock chamber under inert gas atmosphere by means
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5 of an air-tight transfer vessel.
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8 XPS measurements were conducted in an ultrahigh vacuum surface analysis system by
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10 monochromatized Al K α (1486.6 eV) radiation (PHI 5800 MultiTechnique ESCA System,
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12 Physical Electronics) at a base pressure of 10⁻¹⁰ mbar. During data acquisition of survey and
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14 detail spectra, the constant-analyzer-energy mode was used at 93.9 and 29.35 eV energy
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16 passes, respectively. For elemental depth profiling, the outer surface layers were removed by
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18 Ar⁺ sputtering (ISp $\sim$ 1 mA; USp = 5 kV) for 10 minutes. Acquired XP spectra were
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20 analysed by CasaXPS software package (Casa Software) and the main C 1s peak was set to
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22 284.6 eV (C-C/C-H bond in carbon super C 45³²) for all the spectra except for Na-CMC
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24 reference spectra which has been calibrated at 285 eV (hydrocarbon contaminations,
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26 adventitious carbon^{33,34}).
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31 All the spectra presented in the manuscript were acquired under the same conditions. In order
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33 to enable direct comparison of the same core level region for different samples all spectra
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35 were normalized. Accordingly, the peak area for the sample exhibiting lower intensity (peak
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37 area) for all detected regions were multiplied by a normalization factor. The normalization
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39 factor for each sample (Y_i) has been obtained by dividing the ' $SumRef$ ' to ' $SumY_i$ '. Herein,
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41 ' $SumY_i$ ' indicates the sum of the 'peak areas in each core level divided by their sensitivity
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43 factor' for a specific sample, while ' $SumRef$ ' is that of the sample with the highest detected
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45 peak areas in all regions. The secondary electron background of each spectrum is subtracted
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47 by a Shirley background and the spectra are fitted employing a combination of Gaussian and
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49 Lorentzian line shape at a ratio of 70:30. The FWHM has not been constrained during the
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51 fitting process. Overall the difference between FWHM of different peaks of the same
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53 component varies up to 0.6 eV.
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The reproducibility of the results was checked by performing the analyses on at least two electrodes for different conditions. As an example, **Figure S1** and **S2** in the Supplementary Information section report duplicate results for the electrode in contact with the electrolyte (LP30) at OCV and the cycled electrode in the lithiated state (2nd discharge at 2.5 V).

The calculated atomic concentrations of the CEI layer species have been reported in arbitrary units, while the calculated percentage values are reported in **Table S1** in the Supplementary Information section.

3. Results and discussions

An accurate identification of the chemical species belonging to the investigated electrode material and its interface with the electrolyte solution is crucial for a better understanding of its properties and electrochemical behaviour. XPS is a powerful tool and widely used technique that can provide chemical and quantitative information on the studied systems. Prior to start the investigation on the LR-NMC system, a careful review of the main components associated with the EEI layer on LR-NMC previously reported in literature has been performed.

Table S2 in the Supporting Information section presents a summary of the binding energies of different components reported in literature. Combining these literature values with those obtained for the reference materials used in this work, see **Table 1**, allows us performing a careful and accurate assignment of the XPS peaks obtained for the investigated electrodes.

Table 1

3.1. XPS of reference materials

XPS analysis was firstly conducted separately on individual constituents of the cathode material, *i.e.* active material (LR-NMC), carbon additive (super C 45) and binder (Na-CMC) and on their combination (pristine composite LR-NMC electrode). The results represent a useful dataset for the analysis of the XPS spectra of the whole electrodes allowing to investigate the origin of the EEI film formation and to distinguish the contribution of each component material.

While the XP spectra of the reference materials are not reported here, **Table 1** reports the peaks assignment in the Li 1s, C 1s, O 1s, F 1s, Na 1s and Mn 3p region for the investigated compounds. The main carbon feature in Carbon super C 45 is calibrated to 284.6 eV which is in agreement with Yi-Chun Lu *et al.*³² observations for carbon black. The spectra shows one asymmetric peak at 284.6 eV with a long tail at higher energy introducing a peak at ~290.5 eV known as the shake-up satellite structure ($\pi \rightarrow \pi^*$ transition) of carbon black.

In the C 1s core level spectrum of Na-CMC, four different peaks were detected. The first peak at lowest binding energy is calibrated to 285 eV and it is attributed to the C-C component. The three other peaks at higher binding energies, *i.e.* 286.6 eV, 288.4 eV and 289.9 eV, are attributed to C-O, O-C=O and CO_3^{2-} groups, respectively.^{33,34}

In order to evaluate the chemical reactivity of the electrolyte with the pristine LR-NMC electrode in OCV condition, Li/LR-NMC half-cells with LP30 electrolyte were prepared and stored for 24 h before being analysed by XPS. The analysis allows for a further distinction between chemical and electrochemical reactions contributing to EEI film formation. The obtained binding energy values for the separate components of the composite electrodes have been reported in Table 1, while **Figure 1** exhibits the XP core level spectra of Li 1s, O 1s, C 1s and F 1s regions of LR-NMC cathode materials as pristine powder, composite electrode and composite electrode in contact with electrolyte.

Figure 1

In the Mn 3p region, the identified peaks are assigned to Mn^{4+} ³⁵ which is in agreement with the expected Mn oxidation state in LR-NMC materials. In the O 1s region, two peaks are mainly observed which are attributed to the lattice oxygen³⁶ (at lower binding energy side) and the oxygen bonded to the carbon by a sigma bond (at higher binding energy side). For the pristine LR-NMC powder, an extra peak between these two is resolvable which can be related to the presence of lower coordinated oxygen ions at the sample's surface.³⁷ Accordingly, in the Li 1s region, the peak observed is ascribed to the Li present in the cathode structure.

The XP spectra of the composite cathode material stored for 24 hours in contact with the electrolyte (LP30) inside a cell exhibits a chemical composition similar to the pristine composite cathode. However, it is worth noting that the latter one exhibits a new peak in the Li 1s region ascribed to the presence of LiF. Additionally, three extra peaks associated with the formation of LiF, $\text{Li}_x\text{PO}_y\text{F}_z$ and $\text{Li}_x\text{PF}_y/\text{LiPF}_6$ are detected in the F 1s regions.

The observation is in agreement with the formation of lithiated fluorophosphates reported for LiCoO_2 soaked in electrolyte detected both at the F 1s and O 1s region.³⁸

These new peaks account for chemical reactions occurring at the EEI. Indeed, the presence of transition metal within the layered structure triggers decomposition processes at the EEI, even when no current is applied to the cell,³⁹ by forming a thin layer on the electrode's surface. Additionally, as also previously reported,^{40–43} hydrolysis of LiPF_6 salt by traces of water inside the electrolyte can further accelerate salt decomposition and formation of hydrofluoric acid (HF) and fluorophosphate-based species as summarized in the reaction scheme (I) reported below.



3.2. XPS of electrochemically cycled cells

3.2.1. Comparison of binder-free and composite electrodes

The effect of the Na-CMC binder on the chemical composition of the EEI layer has been investigated to better understand the reactivity of the LR-NMC active material toward the electrolyte. The comparison has been performed between the binder-free electrode (85 wt% LR-NMC active material and 15 wt% conductive carbon super C 45) and the composite electrode (85 wt% LR-NMC, 10 wt% super C 45 and 5 wt% Na-CMC binder). For this purpose, cells were cycled within the 2.5 - 4.8 V voltage range for two cycles and then disassembled in the lithiated state. The voltage profiles are showed in **Figure S3** in the Supplementary Information section. The electrodes recovered from the cells were subjected to a rinsing step to remove electrolyte residuals. The XP core level spectra obtained in the Li 1s - Mn 3p, C 1s, O 1s and F 1s regions of the investigated electrodes are shown in **Figure 2**.

Figure 2

In the C 1s region, the peak associated with the C-C bond containing species is more pronounced for the binder-free electrode compared to the composite one. This can be

explained by taking in consideration the different composition of the two electrodes, *i.e.* higher amount of carbon Super C 45 in the binder-free electrode.

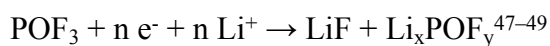
To get insights on the thickness of the surface layer formed on the two electrodes, we considered the intensity of the Mn peak as an indicator of the amount of products formed at the EEI layer. Indeed, a thicker layer induces a screening effect on the Mn XPS signals as a consequence of the detection depth limit. Accordingly, we can reasonably claim that the EEI layer on the binder-free electrode is thicker than the one formed on the composite electrode. The observation is further confirmed by the intensity of the peaks ascribed to electrolyte decomposition products in the C 1s and F 1s region. Indeed, the peaks are more pronounced for the binder-free electrode, indicating a decreased reactivity of the composite electrode toward the electrolyte. This observation highlights the critical importance of the electrode processing and composition for an improved stability of the EEI. Indeed, the observation indicates that the presence of Na-CMC binder in a composite electrode mitigates the catalytic activity of the active material toward the electrolyte.⁴⁵ Na-CMC constitutes not only an optimal binder agent but acts as a stabilizing agent by covering and protecting the electrode material. It is worth mentioning that the environmental benignity of LIBs components is a high priority target in view of their upscale for large scale production. Herein, we demonstrate that aqueous electrode processing employing Na-CMC binder is a valid strategy toward the challenge further enabling a mitigated surface reactivity of LR-NMC with the electrolyte.

It has been previously reported that Na-CMC exhibits a specific reactivity with organic electrolyte leading to the formation of new chemical species after simple contact of the electrode with the electrolyte.⁴⁶ Our results suggest that the CMC reactivity contributes to a passivation layer formation on the cathode's surface that suppresses the electrolyte

decomposition upon cycling, as demonstrated by the inferior amount of electrolyte decomposition products found on the composite cathode.

Figure 2 (e) and (f) report the calculated atomic concentration of carbonaceous and fluorine-containing species respectively, for composite and binder-free electrodes. Interestingly, the atomic concentration of carbonaceous-based products is not affected considerably implying that solvents degrade to a similar extent once in contact with both electrodes. On the other hand, the increased amount of salt decomposition products (represented in fluorine containing region) for the binder-free cathode compared to the composite electrode suggests an increased decomposition rate for LiPF_6 when directly in contact with the binder-free cathode, thus suggesting an enhanced reactivity of the LR-NMC active material when not in presence of Na-CMC, displayed as increased catalytic activity of the transition metals and oxygen in the LR-NMC lattice.

LiF results to be the dominant compound, however, due to the reactivity of LiPF_6 salt with residual water traces in the electrolyte the following interaction can occur resulting also in the formation of phosphates and fluoro-phosphates species as described in the reaction scheme (II) reported below.



The lower surface area of the composite electrode conferred by the presence of Na-CMC is most likely responsible of the decreased reactivity toward the salt. However, it is worth mentioning that LiPF_6 dissociation produces PF_5 which is a strong Lewis acid. The latter one

can further react with the hydroxyl groups in Na-CMC binder leading to the formation of POF-like compounds as in the reaction mechanism (III) described below:



This reaction path could explain the higher ratio of $\text{Li}_x\text{PO}_y\text{F}_z$ compared to LiF in the composite electrode. Indeed, the presence of the binder and its hydroxyl groups catalyses reaction (III) thus producing much more $\text{Li}_x\text{PO}_y\text{F}_z$ than LiF. On the other hand in the binder-free electrode reaction (II) dominates leading to the formation of $\text{Li}_x\text{PO}_y\text{F}_z$ and LiF in comparable amount.

We can conclude that the accelerated salt decomposition process induced by Na-CMC and the one activated by the catalytic activity of the transition metals in the cathode active material lead to the formation of phosphates and fluoro-phosphates. However, the kinetics of the salt decomposition in contact with the cathode active material are much faster (thus producing much more compounds) than the one characterizing the same process in presence of Na-CMC, thus explaining the larger amount of fluorinated species in the binder-free electrode.

3.2.2. Composite electrodes

3.2.2.1 The upper cut-off voltage: de-lithiation effect on EEI layer formation

Figure 3

The investigation of the effect of the de-lithiation voltage on the EEI film formation in LR-NMC has been performed by cycling cells within different potential windows.

Figure 3 reports the corresponding voltage profiles of cells cycled at increasing upper cut-off voltages, *i.e.* 4.2, 4.4, 4.6 and 4.8 V and discharged down to 2.5 V. The electrodes were recovered for post-mortem XPS analysis in the de-lithiated state and the corresponding XPS analysis is reported in **Figure 4** showing the XP core level spectra of the Li 1s – Mn 3p, C 1s, O 1s, F 1s and P 2p regions. For comparison purposes, the analysis has been performed after the third de-lithiation process for all the samples as reported in Fig. 3. Indeed, when cycling above 4.4 V during the first two cycles two concomitant processes occur, namely the activation of the LR-NMC cathode and the formation of the CEI layer. To deconvolute these two effects, the investigation has been carried out when the activation process is concluded and the chemical composition of the CEI layer is not affected by it.

Figure 4

For all the examined electrodes, the Li 1s region, exhibits a small contribute at about 54.2 eV associated with the detection of lithium in the lattice of LR-NMC. As expected, the peak intensity is decreasing with increasing of the upper cut-off voltage as a consequence of the higher amount of lithium removed from the cathode's structure.

The peak intensity associated with the detection of Mn and M-O observed in the Mn 3p and O 1s regions is also decreasing with an increased de-lithiation degree, especially for the electrodes charged up to 4.6 V and 4.8 V. The Co and Mn peaks in the Co 2p and Mn 2p regions, which are not shown here, present the same trend. This can be explained considering the formation of a thicker layer of decomposition products on the electrode's surface, reflecting an enhanced oxidation of the electrolyte. Taking into account that the XPS depth penetration is approximately less than 10 nm, the detection of the Mn peaks in all cycled

samples indicates that the EEI layer on LR-NMC samples present a thickness that does not exceed 10 nm.

For the electrode de-lithiated up to 4.2 V, the detection of carbonate species in the O 1s region indicates the presence of Li_2CO_3 , produced by electrolyte decomposition and nucleophilic attacks of superoxide species to carbonate solvents.

Interestingly, it is observed that this feature decreases with an increased degree of de-lithiation. This can be explained by two possible mechanisms. The first one involves the electrochemical oxidation of Li_2CO_3 , which is unstable at potential values higher than 4.2 V²³. The second one is related to the reactivity of the LiPF_6 salt toward the formed Li_2CO_3 . Indeed, GC-MS studies⁴⁹ have reported the formation of CO_2 and POF_3 gases, according to the reaction path (IV):



This reaction is in agreement with our observations. Indeed, while the Li_2CO_3 peak is decreasing, the LiF signal in the F 1s region is increasing.

Additionally, the P 2p region highlights an increased amount of $\text{Li}_x\text{PO}_y\text{F}_z$ species (peak areas increase), suggesting also an electrochemical activity of the generated gaseous species (POF_3) according to the reaction (V) reported below.



Additionally, it should be considered that the increased detection of LiF can be explained considering the reactivity of LiPF_6 with radical species formed up to 4.2 V developing more

HF and POF_3 species, and subsequently inducing LiF and salt decomposition products formation in an autocatalytic reaction as described by A. Guéguen *et al.*⁵⁰

The overall increase of LiF in the EEI layer is clearly showed in Figure 4 (g) reporting the calculated atomic concentration of fluorinated species as calculated from the XP spectra. It is worth noting that an opposite trend is observed for the $\text{Li}_x\text{PO}_y\text{F}_z$ contribution in the F 1s and P 2p regions. This can be explained considering the different information depth of XPS in the F 1s and P 2p regions. Indeed, electrons ejecting from F and P have around 500 eV difference in kinetic energy, which according to the universal curve,⁵¹ results in 0.9 nm difference in their inelastic mean free path.

The negligible changes observed in the organic components over increasing cut-off voltage values (see Figure 4 (f) reporting the atomic concentration of the carbonaceous species), indicates a stable organic component of the layer, which is formed during the first cycles and it is stable upon cycling, as indicated by the constant area of the PEO species deriving from EC polymerization and C-C contribution in the C 1s region.

3.2.2.2. EEI layer composition in the de-lithiated and lithiated state

To shed light on the stability of the EEI layer formed after the activation voltage plateau and its chemical composition upon lithiation, further investigations have been carried out on electrodes recovered in the lithiated state (at 2.5 V) after being charged up to 4.8 V.

The corresponding XP spectra of the Li 1s, C 1s, O 1s and F 1s regions, for the charged electrode (3rd charge at 4.8 V) and the discharged electrode (2nd discharge at 2.5V), are shown in **Figure 5**.

Figure 5

The features associated with the detection of Mn, C-C and M-O groups are more pronounced for the de-lithiated electrode (4.8 V) when compared to the lithiated electrode (2.5 V). This indicates a thinner EEI layer formed in the de-lithiated state. Indeed, in the lithiated state the intensity of such contributions is reduced due to the presence of a thicker film on the electrode's surface. It is worth noting that the XPS contributions related to the EEI layer components of the lithiated electrode appear at slightly higher binding energies, suggesting a charging effect responsible for the shift. This can be explained considering the chemical and morphological nature of the EEI layer, which accordingly is thicker and less conductive than the one obtained after de-lithiation.⁵²

The F 1s region gives some insights on the observation, especially when comparing the intensity of the peaks related to the presence of LiF.

According to the calculated atomic concentration reported in Figure 5 (f) the amount of LiF component exceeds for the de-lithiated electrode. To explain the higher amount of LiF in the de-lithiated state, we could think at the EEI layer as the mosaic model of the SEI at the anode side proposed by Peled,⁵³ where the inner layer is composed by inorganic components and the outer layer is mostly organic based. Based on this assumption, the higher detection of LiF in the de-lithiated state might also derive from the presence of a thinner layer of organic components on top of the inner layer of the EEI film, thus enabling an increased detection of LiF.

However, this model does not describe properly the EEI layer in this work for two main reasons. First, the organic component appears to be very stable both in the lithiated and de-lithiated states, thus the increased detection of LiF cannot be explained considering the outer layer being mainly organic. Second, if inorganic compounds are supposed to be detected in the deeper EEI layer, the increased amount of $\text{Li}_x\text{PO}_y\text{F}_z$ in the lithiated state should not be

observed as, indeed, it is very evident in Fig. 5 (d) in the F 1s region. The observation can be instead explained considering that $\text{Li}_x\text{PO}_y\text{F}_z$ components are less stable at high voltages and can undergo electrochemical oxidation losing their contribution to EEI film to some extent.⁵⁴ Alternatively, it is well known that $\text{Li}_x\text{PO}_y\text{F}_z$ species are generated upon reduction of POF_3 (see reaction (V)), which is clearly more pronounced in the discharged state at 2.5 V. Thus, the higher amount of LiF in the de-lithiated state can be associated with the activity of superoxide species toward carbonate solvents, leading to the formation of Li_2CO_3 , which can further react with LiPF_6 and generate POF_3 , CO_2 and LiF. This would explain the increased amount of LiF and the decreased amount of Li_2CO_3 in the de-lithiated state.

3.2.2.3. Effect of tris(pentafluorophenyl)borane (TPFPB) additive in LR-NMC cells

The formation of a stable surface film on the electrode is critical for the electrochemical performance of LIBs. This applies to both positive and negative electrodes. In the ideal scenario, the surface film should enable the conduction of lithium ions and hinders the electronic flow from the electrodes to the electrolyte thus passivating the electrodes' surface and limiting electrolyte decomposition.

It is well known that inorganic species such as LiF and Li_2CO_3 although necessary for the stability of the SEI and prevention of its continuous growth, in high amounts may lead to the formation of a non-conductive layer hindering lithiation of the electrode and leading to an increased cell resistance.

For instance, perfect LiF crystals (i.e., no defective) present a very limited electronic conductivity in view of their wide band gap, thus blocking further electron flow from the electrode to the electrolyte leading to further electrolyte decomposition. On the other hand, theoretical calculation showed that the ionic conductivity for LiF is approximately 10^{-31} S

cm⁻¹ when in contact with a negative electrode and about 10⁻¹² S cm⁻¹ on a positive electrode.⁵⁵

SEI modifiers, in the form of additives, such as tris(pentafluorophenyl)borane (TPFPB), are able to re-dissolve the inorganic SEI components (*e.g.*, LiF, Li₂CO₃, Li₂O)⁵⁶⁻⁵⁹.

TPFPB is an electrolyte additive known to improve the resistance at the LiCoO₂/electrolyte interface⁶⁰ and to modify the interface for Li-rich, Mn-rich (LMR) layered composites.⁶¹

In addition, it is well known that an incomplete dissociation of LiPF₆ can generate ion-paired LiPF₆ more prone to undergo decomposition by forming LiF, PF₅⁶² and HF, which is one of the responsible agents for the EEI layer instability.

The full dissociation of LiPF₆ can be enhanced by Lewis acids because of their ability to form complexes with the PF₆⁻ anions, further hindering HF generation. TPFPB is a strong Lewis acid and anion receptor, thus its use in electrolyte solution effectively hinders the formation of PF₅ and HF by forming complexes with PF₆⁻ anions.⁶²

In this work, a 0.2 molar TPFPB in LP30 solution was used in comparison to the bare LP30 electrolyte. The voltage profile of the LR-NMC cells using the two electrolytes is shown in **Figure 6**.

Figure 6

The XPS comparative analysis conducted on the de-lithiated electrodes is shown in **Figure 7**.

Figure 7

In light of what mentioned above, by analysing the surface film composition we expect a decreased amount of LiF and a consequent decrease of the layer thickness.

However, we detect a thicker layer for the electrode cycled in TPFPB-based electrolyte as suggested by the decreased areas and intensities of Mn 3p, C-C and M-O peaks. Furthermore, an increase in PEO content is exhibited by the electrode cycled in TPFPB-based electrolyte (see C 1s and O 1s spectra). On the other hand, a lower LiF content indicates the beneficial effect of TPFPB additive to prevent an excessive growth of highly resistive LiF.⁶³

The trend of the amount of components in the two different EEI layers is also reported in Figure 7 (e, f) showing the calculated concentration of carbonaceous and fluorine-containing compounds. It is schematically shown that when TPFPB additive is employed, more organic species, Li_xPF_y and $\text{Li}_x\text{PO}_y\text{F}_z$ species are detected, while LiF is present in an inferior amount. Indeed, TPFPB, being a strong anion receptor, can break the Li-F bond according to the reaction (VI):



While we confirm the beneficial effect of TPFPB in suppressing excess LiF formation, we also surprisingly detected a thicker EEI layer in the presence of TPFPB additive, while it is assumed by Zheng *et al.* that the improved performance of Li-rich, Mn-rich (LRMR) layered cathode material in the presence of TPFPB is due to its ability to partially re-dissolve LiF or Li_2O , thus possibly leading to a thinner EEI layer.⁶¹ Additionally, it has been reported that $\text{Li}_x\text{PO}_y\text{F}_z$ species are less stable at high voltages due to the oxygen radical evolution and more HF attack,⁵⁴ however, in this work the effectiveness of TPFPB in scavenging the radicals and aggressive acids is proved by keeping the $\text{Li}_x\text{PO}_y\text{F}_z$ concentration higher than when no additive has been used. Beside the scavenging effect, it has also been reported that TPFPB actively participate in the film formation when used with LiFePO_4 electrodes.⁶⁴

To prove its contribution to the layer composition by formation of boron-containing compounds, the B 1s region should be investigated. The presence of the latter ones could account for the thicker layer formed on the electrode cycled in TFPFB-based electrolyte. Despite the B 1s region shows very low intensity and overlaps with the P 2s region, a larger peak is detected for the electrode cycled with the additive, suggesting the possible presence of borate groups at ~ 193 eV. A comparison of the survey spectra obtained from the electrode cycled in bare LP30 and TFPFB added electrolyte is reported in **Figure S4** in the Supporting Information section.

The presence of boron in the EEI layer, which has been already reported for high voltage cathodes,⁶⁵ can be linked to the thicker EEI layer detected in presence of TFPFB, however further studies are required to confirm its presence. Another argumentation in support of the thicker layer can be derived by considering that in presence of TFPFB the chemical composition of the passivation layer changes, as confirmed by the inferior amount of LiF and increased concentration of organic PEO oligomeric species and $\text{Li}_x\text{PO}_y\text{F}_z$ components, which may contribute to the increased thickness of the layer and may also be beneficial for the cell performance as demonstrated by the voltage profile in Figure 6, especially in terms of improved delivered capacity.

3.2.2.4. Effect of cycling temperature

Initial EEI formation temperature could play a substantial role in the stability of electrode/electrolyte interface and the cycling behaviour, especially in terms of capacity retention. Aurbach *et al.* reported that surface films formed at high temperatures remained stable and did not change at lower temperatures for $\text{LiMn}_{1.5}\text{Ni}_{0.5}\text{O}_4$ electrodes. They

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3 attributed the unique stability to the presence of Ni ions in the lattice. Indeed, as a matter of
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5 fact LiMn_2O_4 (spinel structure) electrodes are extremely unstable at elevated temperatures.⁶⁶
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10 In this work, the effect of the temperature on the chemical composition of the LR-
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12 NMC/electrolyte interface has been investigated at two different temperatures, *i.e.* 20 °C and
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14 55 °C. The corresponding voltage profiles are reported in **Figure 8**.
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17 18 19 **Figure 8**

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24 XPS analysis has been performed on cycled electrodes in the de-lithiated state.
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33 **Figure 9** shows the compared XP core level spectra of the Li 1s, O 1s, C 1s and F 1s regions
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35 for the electrodes cycled at room temperature (20 °C) and at 55 °C. The intensity related to
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37 the M-O, Mn and C-C peaks clearly indicate stronger electrolyte degradation at high
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39 temperature which results in a thicker EEI layer for the electrode cycled at 55 °C. In spite of
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41 some reports confirming the breakdown of the anode SEI at high temperatures, our
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43 observations instead confirm a thicker layer on the cathode's surface as a results of the fast
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45 decomposition kinetics occurring at elevated temperatures.⁶⁷ The F 1s, Li 1s and O 1s
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47 regions, clearly indicate that electrolyte degradation is exacerbated at high temperature
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49 resulting in substantial formation of LiF , $\text{Li}_x\text{PO}_y\text{F}_z$ and Li_2CO_3 species.
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53 In addition, the calculated concentration of the EEI layer species reported in Figure 9 (e, f)
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55 clearly show that contrary to the slight loss of carbonaceous components, substantial increase
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of fluorine containing compounds are observed for the EEIs formed at 55 °C. It is known that the ring opening polymerization of EC is an endothermic process^{68,69} and one could assume that it will result in more PEO oligomeric EEI components at higher temperatures. On the other hand the formed polymeric species can react with PF₅ to form products that may dissolve in the electrolyte or participate in EEI formation.⁷⁰ Therefore, salt decomposition reactions that are kinetically more favourable at high temperatures result in an increased rate formation of the salt decomposition products and react with PEO oligomers resulting in the generation of high amounts of LiF and Li_xPO_yF_z-like components.

Furthermore, it is worth mentioning the study of K. Edström *et al.*⁶⁷ reporting that the F-containing components amount decrease by increasing the EEI formation temperature, might be associated with the screening effect exerted by a layer of organic component overlying the fluorine-containing species. However, based on our in depth XPS profiling studies at room temperature and elevated temperature (see **Figure S5** in the Supporting Information section), a clear distinction of the organic and inorganic layers is not observed on the LR-NMC cathode. Figure S5 shows indeed that before and after sputtering the distribution of organic and inorganic species on the electrode's surface does not exhibit a linear trend, suggesting a non-layered structure of the EEI layer.

4. Conclusions

The XPS analysis performed in this work reveals that Na-CMC binder interacts with the electrolyte solution to form a passivating film that slows down the kinetic of the salt decomposition process, thus stabilizing the EEI layer. By increasing the higher cut-off

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3 voltage, EEI inorganic components grow while the organic components remain stable. The
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5 thinner EEI with higher LiF content observed for de-lithiated state compared to the lithiated
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7 one suggests the decomposition of EEI at high voltages due to the reactivity of superoxide
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9 species with electrolyte and EEI components. The use of TFPB-based electrolyte prevents
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11 excessive LiF formation at high voltages and induces a change in the chemical composition
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13 of the EEI layer, which is characterized by further PEO and $\text{Li}_x\text{PO}_y\text{F}_z$ formation and the
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15 possible presence of boron containing compounds. Finally, fast kinetics at elevated
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17 temperatures exacerbate the salt decomposition which results in an EEI which is thicker and
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19 richer in LiF. The investigation performed in this work suggest a non-layered structure of the
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21 LR-NMC/electrolyte interface layer which cannot be described by the typical mosaic type
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23 model proposed for anode materials and generally accepted also for positive electrodes.
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26 Indeed, upon de-lithiation an increased amount of inorganic LiF is observed when compared
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28 to the lithiated state. According to the mosaic model employed to describe the SEI on
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30 graphite negative electrodes, the increased detection of inorganic components should be
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32 accompanied by the decomposition of the outer layer of organic components in the de-
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34 lithiated state. However, the EEI organic component remains very stable upon cycling of LR-
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36 NMC electrodes, thus excluding the typical SEI model involving the inorganic components to
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38 form the inner part of the SEI. In spite of other reports proposing a reversed structure with F-
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40 containing compounds dominating the outer layer, our in depth XPS study clearly reveals the
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42 non-layered structure of the EEI on LR-NMC electrodes since a clear distinction between
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44 organic and inorganic components occurring at different depths is not observed.
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54 Associated Content

56 Supporting Information

Reproducibility data for the electrode stored in contact with the electrolyte (LP30) at OCV and the cycled electrode in the lithiated state (2nd discharge at 2.5 V). Summary of the atomic concentration percentages of the fluorinated and carbonaceous CEI species detected by changing cycling conditions. XPS binding energies of the various compounds reported in literature as possible EEI layer components. Comparison of the voltage profiles obtained by cycling LR-NMC composite and binder-free electrodes in lithium half cells. Survey XPS spectra for LR-NMC electrodes cycled using the bare (LP30) and the additive-containing (0.2 molar TFPFB in LP30) electrolytes and zoomed region highlighting the B 1s, P 2s and P 2p regions. XPS depth profiling of composite electrode at room temperature (20 °C) and 55 °C.

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Reference compounds	Li 1s	C 1s	O 1s	F 1s	Na 1s	Mn 3p
Csp 45		284.6 (C-C/C-H) 290.4 (satellite, $\pi \rightarrow \pi^*$)				
Na-CMC		285 (C-C/C-H) 286.3 (C-O) 288 (C=O) 289.5 (O-C=O)	532.8 (C-O)		1071.7	
LR-NMC powder	54.4	284.6 (C-C/CH) 286 (C-O) 288.5 (C=O) 290.2 (satellite, $\pi \rightarrow \pi^*$)	529.5 (M-O) 531.5 (C-O)			49.6 (Mn ⁴⁺) 51.8
LR-NMC Composite electrode	55	284.6 (C-C/CH) 285.7 (C-O) 288.3 (C=O) 290 (satellite, $\pi \rightarrow \pi^*$)	529.8 (M-O) 532.4 (C-O)		1072.2	49.8 (Mn ⁴⁺) 52
LR-NMC/LP30 Composite electrode in contact with electrolyte	54.3 56.1(LiF)	284.65 (C-C/C-H) 286 (C-O) 288.5 (C=O) 290.1 (O-C=O) / shake-up satellite	529.7 (M-O) 532.6 (C-O)	685.1 (LiF) 686.4 (Li _x PO _y F _z) 688 (LiPF ₆)		49.8 (Mn ⁴⁺) 52

Table 1. XPS binding energies of reference compounds analyzed in this work.

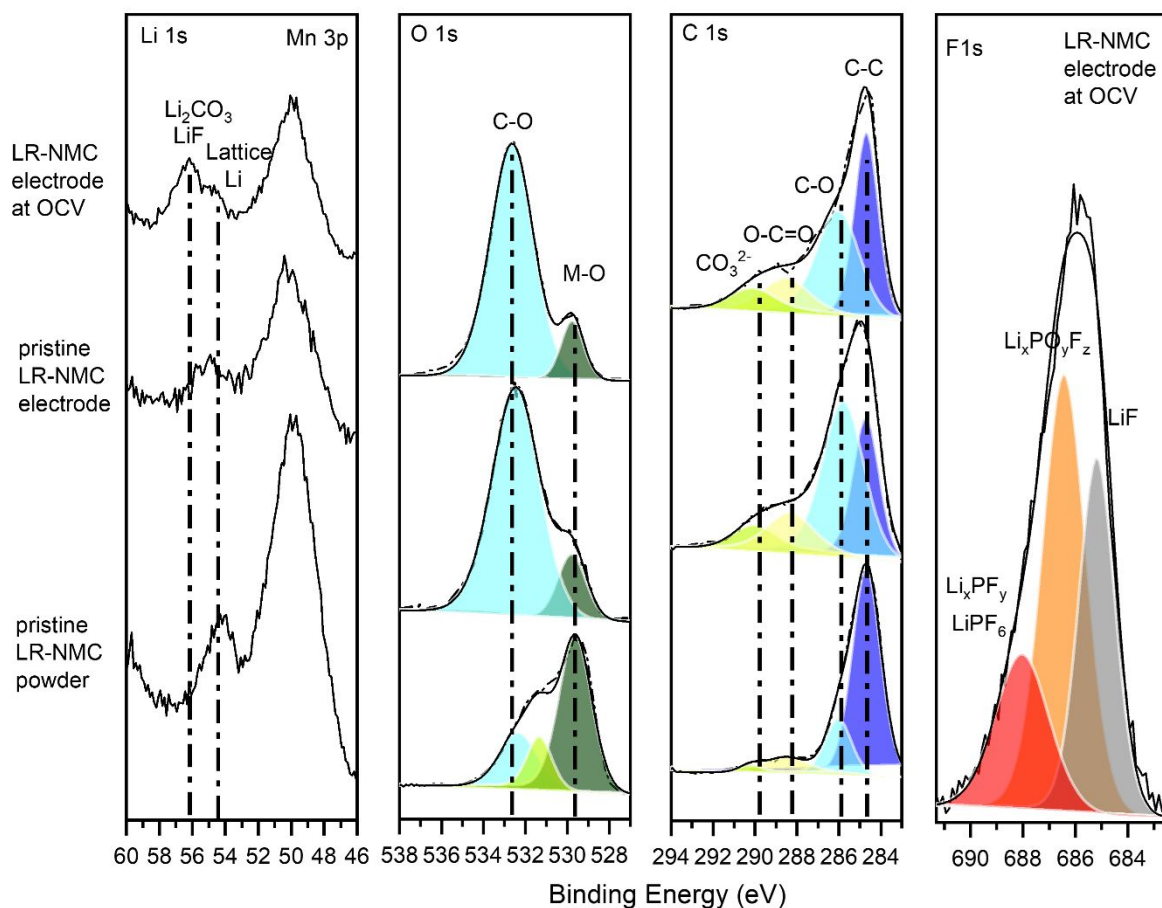


Figure 1. XP core level spectra of the Li 1s, Mn 3p, O 1s, C 1s and F 1s regions of LR-NMC cathode materials as pristine powder, composite electrode and composite electrode in contact with the electrolyte (LP30) at OCV.

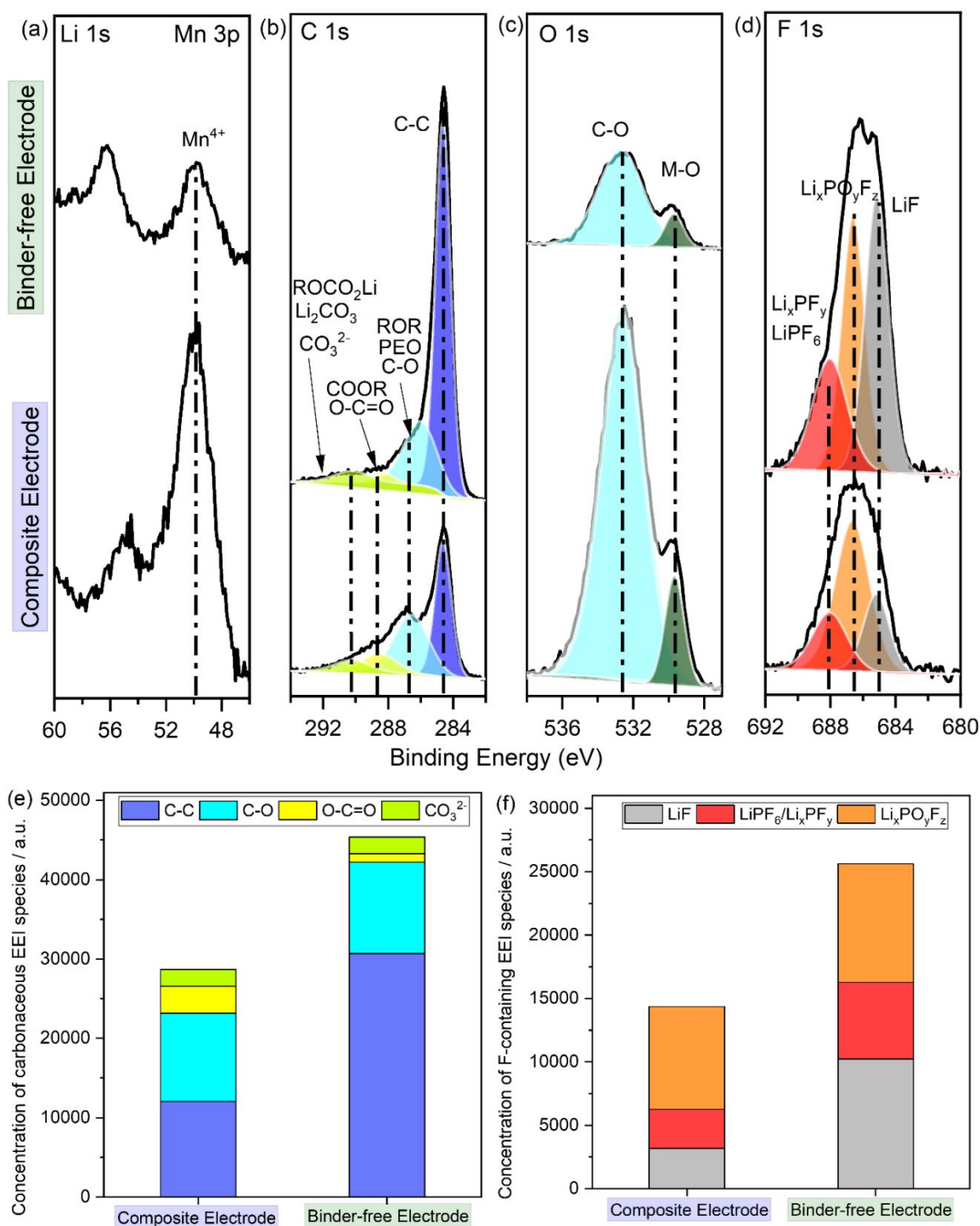


Figure 2. XP core level spectra of the (a) Li 1s, (b) C 1s, (c) O 1s and (d) F 1s regions for both composite and binder-free LR-NMC electrodes. Corresponding atomic concentrations of (e) carbonaceous and (f) fluorine-containing EEI layer species.

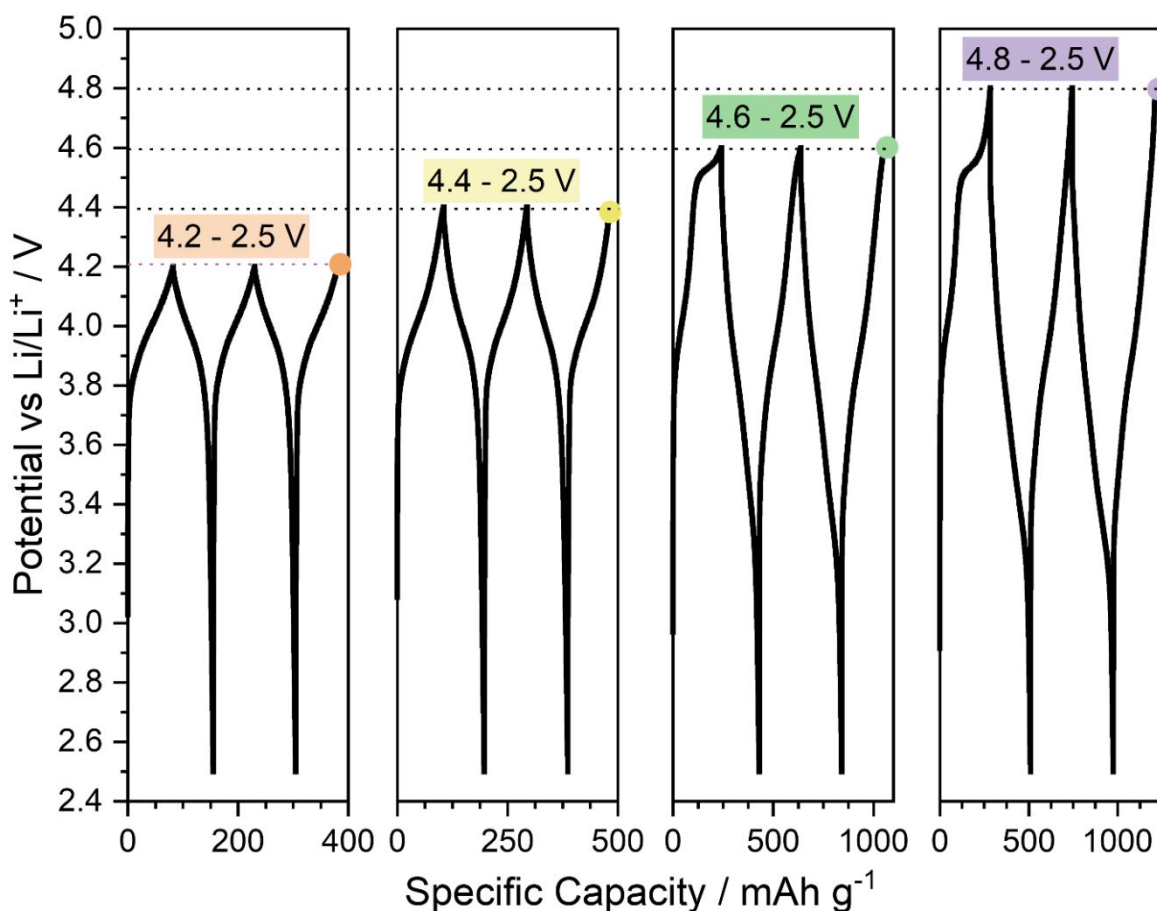


Figure 3. Potential profiles of Li/LR-NMC cells obtained by cycling at C/20 within the 4.2-2.5 V, 4.4-2.5 V, 4.6-2.5 V and 4.8-2.5 V potential regions. Electrolyte: 1 M LiPF₆ in EC:DMC (1:1wt). Temperature: 20 ± 2 °C.

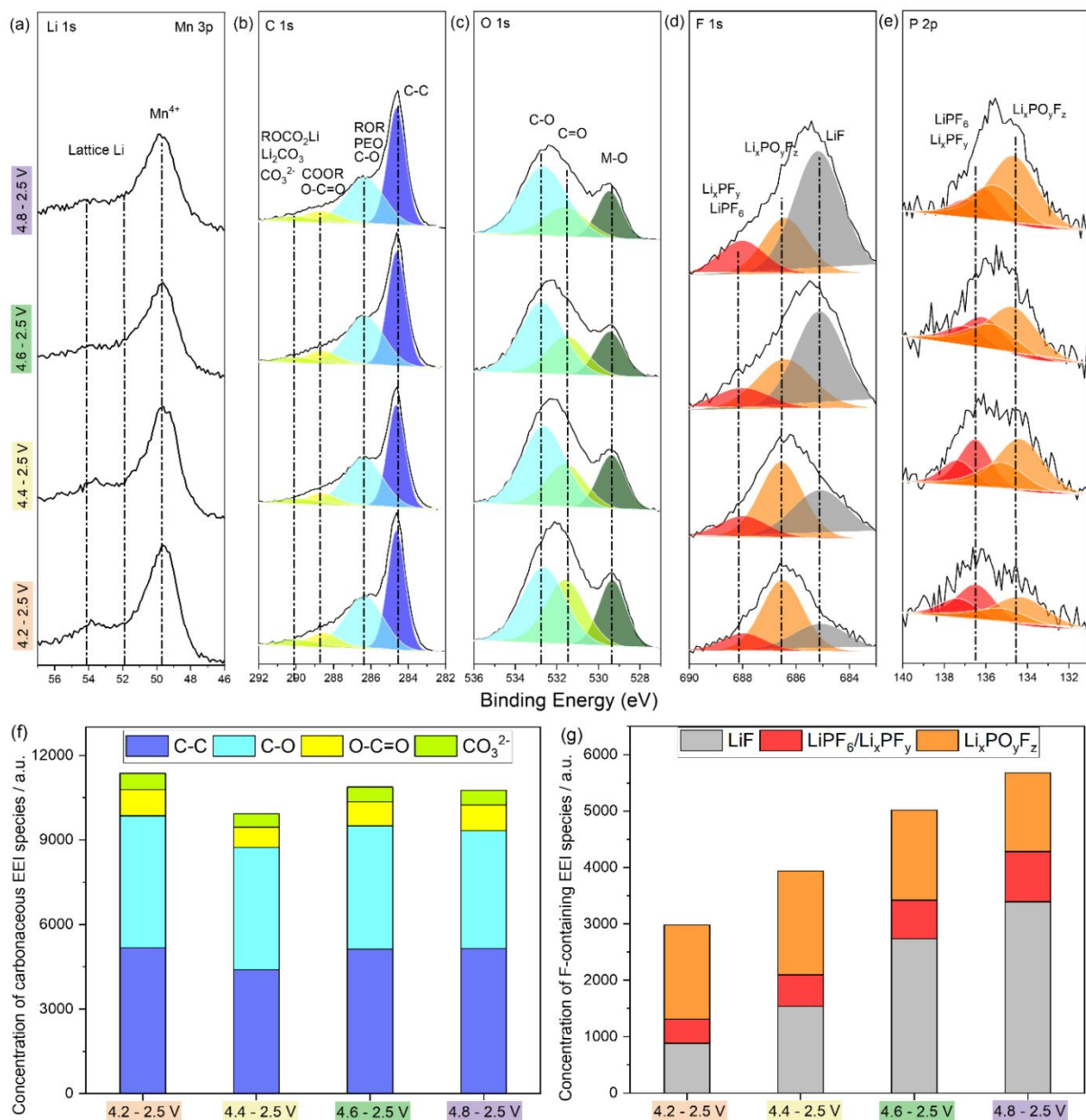


Figure 4. XP core level spectra of the (a) Li 1s, (b) C 1s, (c) O 1s, (d) F 1s and (e) P 2p regions for LR-NMC electrodes at four different de-lithiated states, i.e., after the first charge to 4.2 V, 4.4 V, 4.6 V and 4.8 V. Corresponding calculated atomic concentrations of (f) carbonaceous and (g) fluorine-containing components of the EEI layer species.

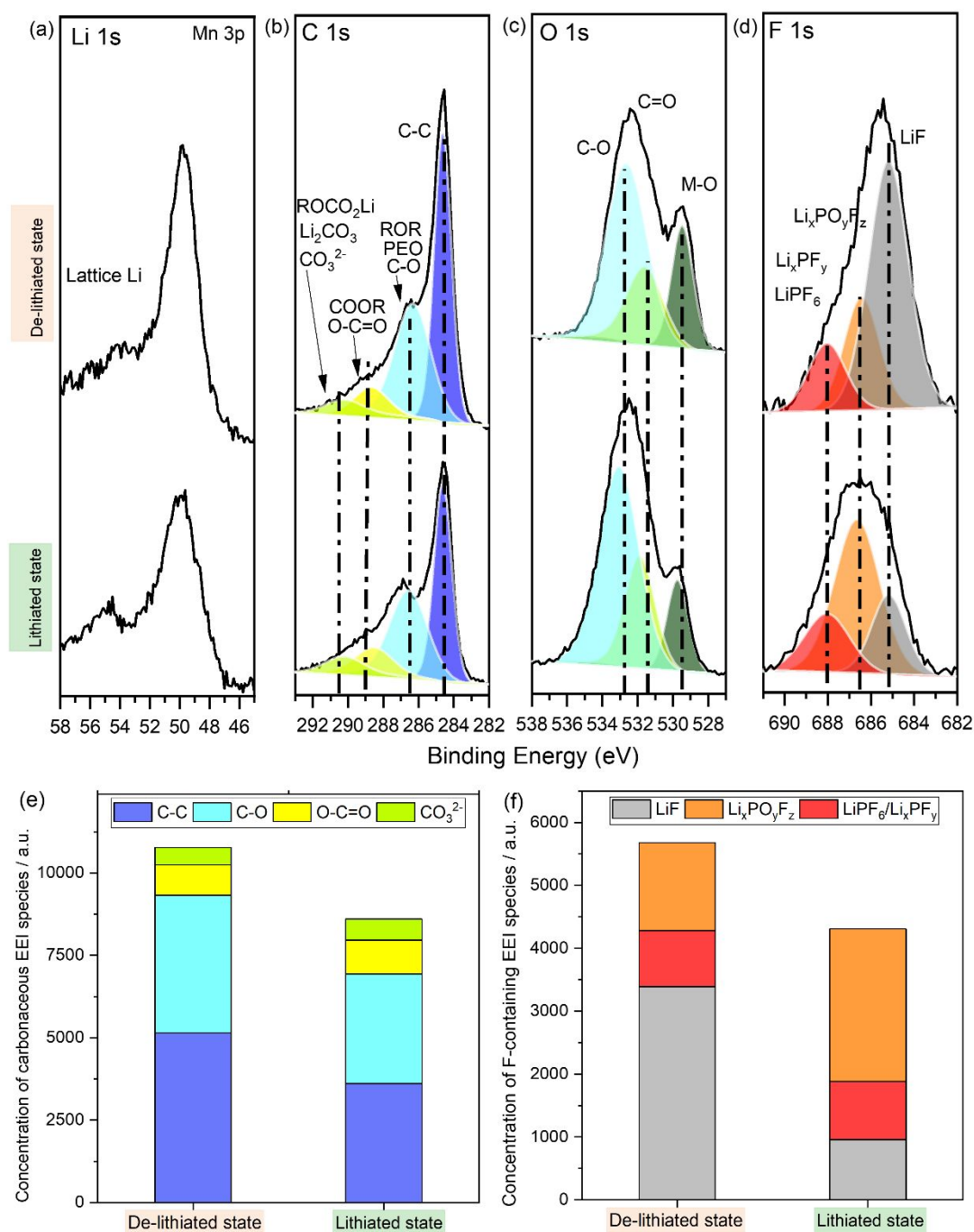


Figure 5. XP core level spectra of the (a) Li 1s, (b) C 1s, (c) O 1s, (d) F 1s regions for LR-NMC electrodes in the lithiated (2nd discharge at 2.5 V) and de-lithiated (3rd charge at 4.8 V) states. Corresponding calculated atomic concentrations of (e) carbonaceous and (f) fluorine-containing components of the EEI layer species.

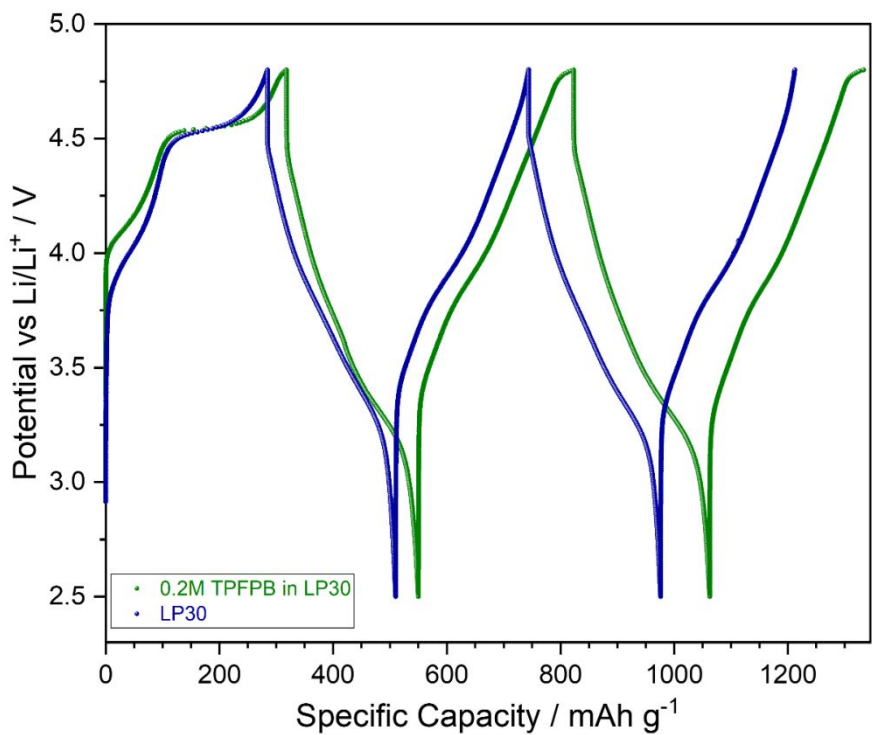


Figure 6. Potential profile of Li/LR-NMC cells obtained by cycling at C/20 within the 2.5-4.8 V potential range by using the bare (LP30, blue curve) and the additive-containing (0.2 molar TFPB in LP30, green curve) electrolytes.

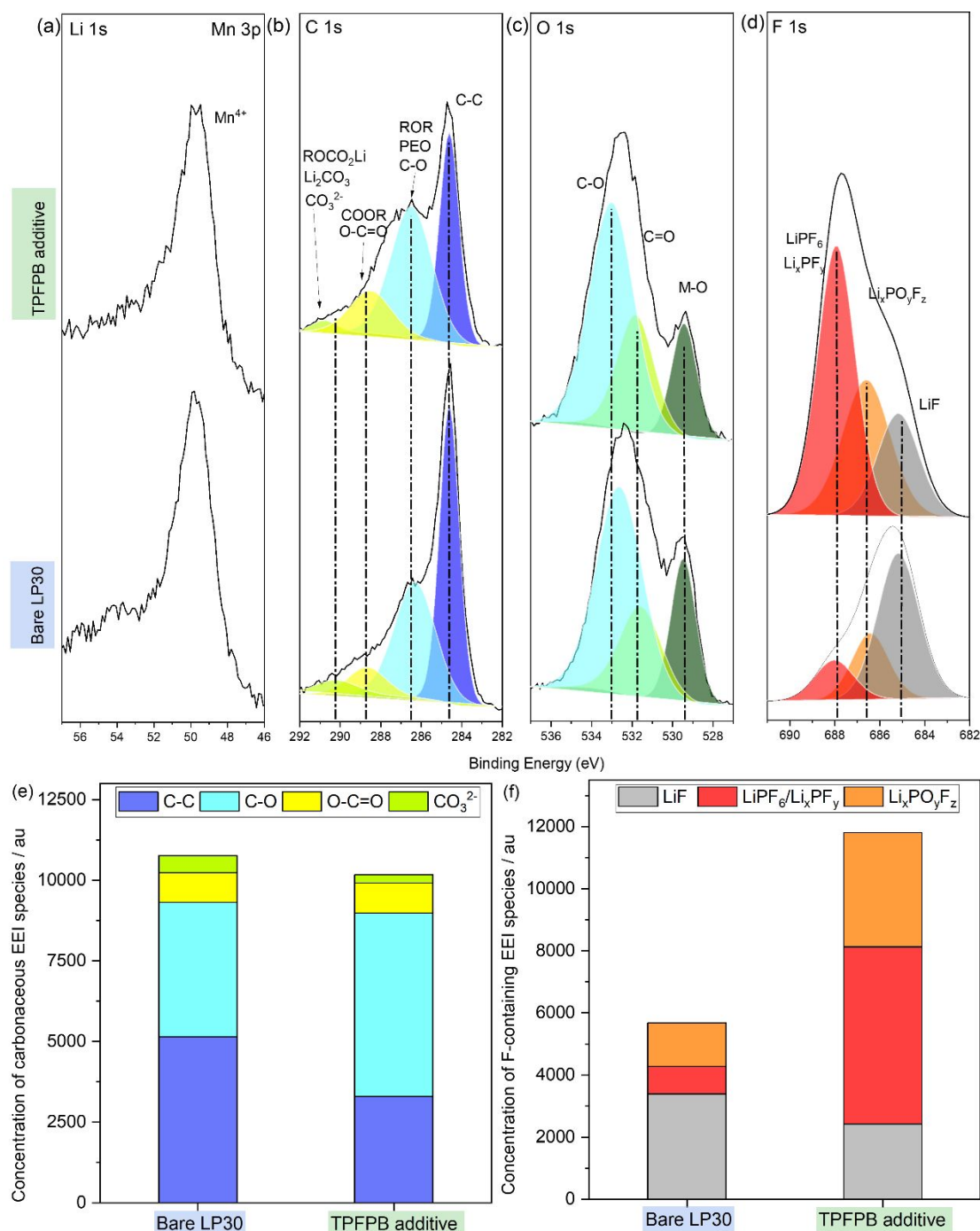


Figure 7. XP core level spectra of the (a) Li 1s, (b) C 1s, (c) O 1s and (d) F 1s regions for LR-NMC electrodes (de-lithiated state, 3rd charge at 4.8 V) cycled using the bare (LP30) and the additive-containing (0.2 molar TFPFB in LP30) electrolytes. Corresponding calculated atomic concentrations of (e) carbonaceous and (f) fluorine-containing components of the EEI layer species.

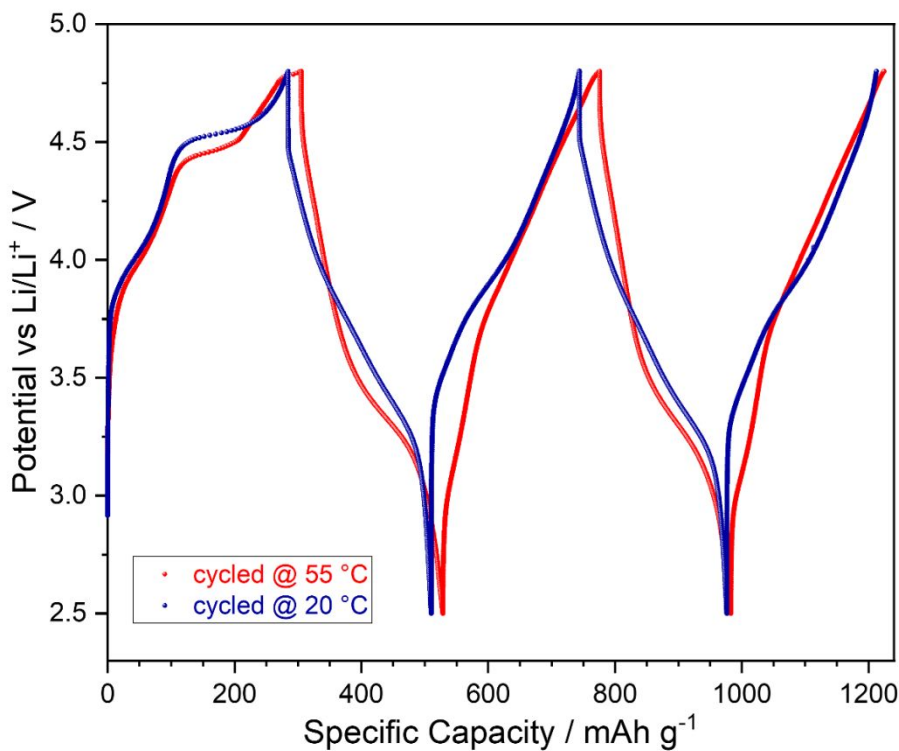


Figure 8. Potential profile of Li/LR-NMC cells obtained by cycling at C/20 within the 2.5-4.8 V potential range at 20 °C (blue curve) and at 55 °C (red curve).

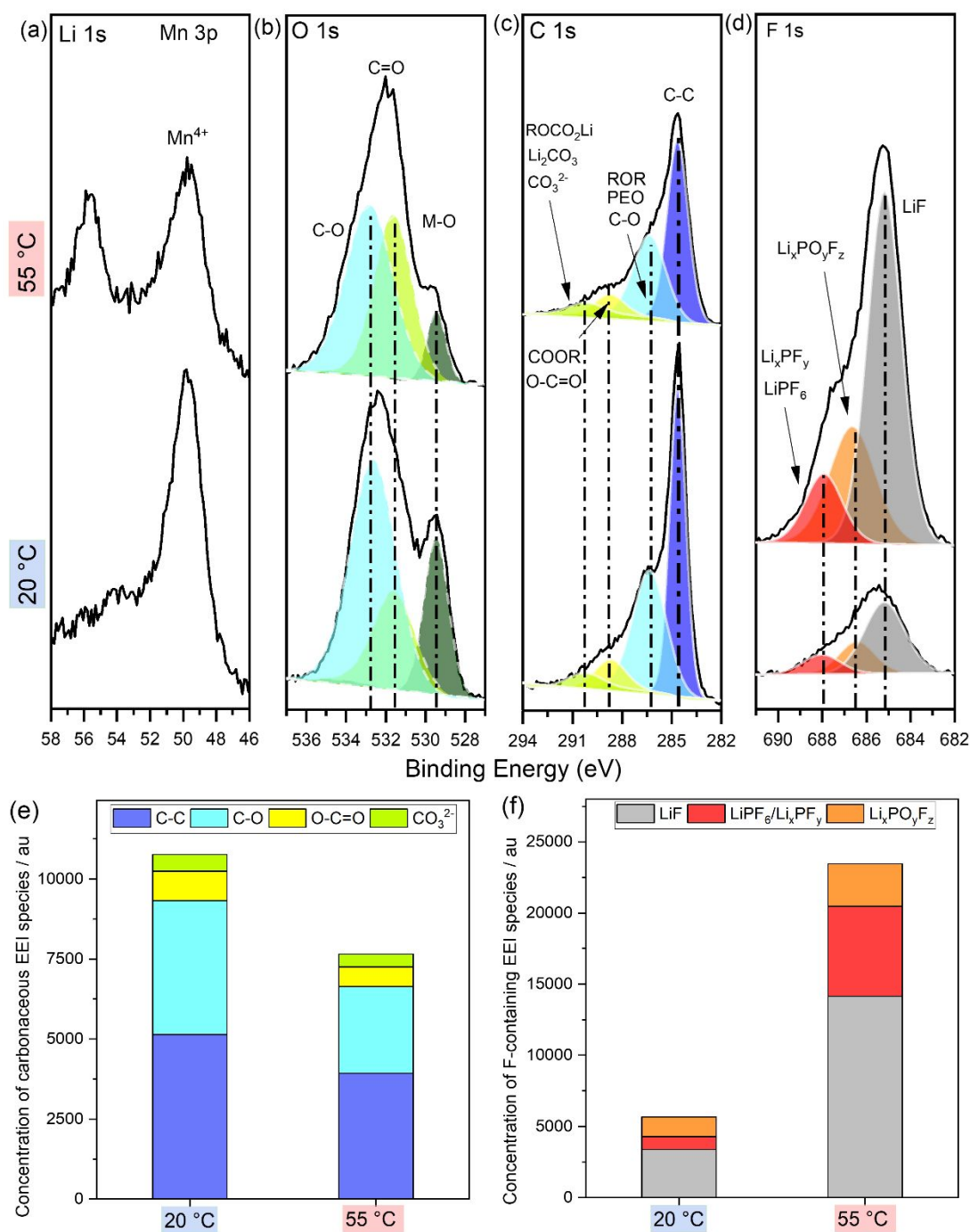


Figure 9. XP core level spectra of the (a) Li 1s, (b) C 1s, (c) O 1s and (d) F 1s regions for LR-NMC electrodes (de-lithiated state, 3rd charge at 4.8 V) cycled at room temperature (20 °C) and at 55 °C. Corresponding calculated atomic concentrations of (e) carbonaceous and (f) fluorine-containing components of the EEI layer species.

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